

7.4 Hydroxy Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.4 Hydroxy Compounds (A Level Only)
Difficulty	Hard

Time allowed: 40

Score: /33

Percentage: /100

Question 1a

This question is about the reactions of phenol and benzene.

Explain the difference in bromination of phenol and benzene. In your answer, include structures of any organic products formed in the reaction.

[7 marks]

Question 1b

A mixture of 2-nitrophenol and 4-nitrophenol can be formed via the pathway shown in Fig. 1.1.

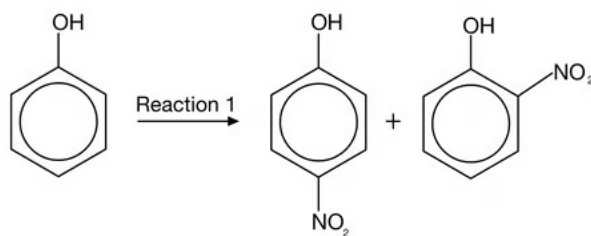


Fig. 1.1

- i)
State the reagents required for reaction 1.

[1]

- ii)
Calculate the mass, in grams, of phenol required to form 30 g of 2-nitrophenol assuming a 35% yield. Give your answer to three significant figures and show your working.

mass = g
[4]

[4 marks]

Question 1c

Phenol reacts with sodium to form a salt and hydrogen gas in an acid-base reaction.

Explain why phenol behaves as a weak acid.

[3 marks]

Question 2a

Phenyl 2-hydroxybenzoate is an antiseptic and is shown in Fig. 2.1.

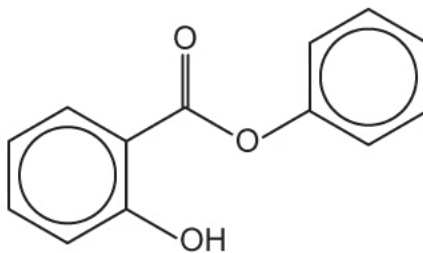


Fig. 2.1

A molecule of phenyl 2-hydroxybenzoate has 13 carbon atoms

i)

State the number of carbon atoms that are sp , sp^2 and sp^3 hybridised carbon atoms, if any, in phenyl 2-hydroxybenzoate.

sp carbons

sp^2 carbons

sp^3 carbons

[1]

ii)

Complete Table 2.1 showing the reactions of phenyl 2-hydroxybenzoate with the three reagents.

Table 2.1

Reagent	Structure of product (s)	Type of reaction
Na		Acid-base reaction
Excess Br_2 (aq)		Bromination / (electrophilic) substitution

[3]

[4 marks]

Question 2b

Phenyl 2-hydroxybenzoate can be made from 2-hydroxybenzoic acid and phenol by heating the two compounds strongly together.

Write an equation, using the displayed formula, for this reaction.

[2 marks]

Question 2c

Complete and balance the equation in Fig. 2.2 for the alkaline hydrolysis of phenyl 2-hydroxybenzoate.

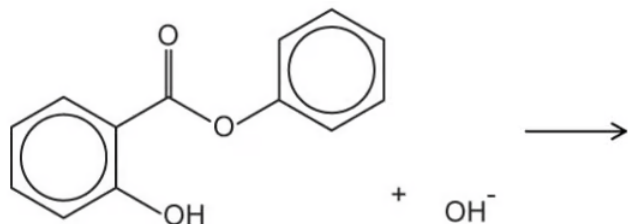


Fig. 2.2

[3 marks]

Question 3a

This question is about a drug called mesalamine (also known as 5-aminosalicylic acid) that is used to treat bowel disease. It can be synthesised from salicylic acid, the chemical structure of which is shown in Fig. 3.1

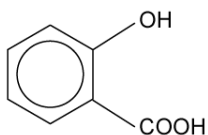


Fig. 3.1

State the systematic IUPAC name for salicylic acid and deduce the **two** positional isomers are possible, giving their names and structures.

IUPAC name of salicylic acid

positional isomer 1	positional isomer 2

[3 marks]

Question 3b

A possible two step synthesis of mesalamine is shown below in Fig. 3.2.

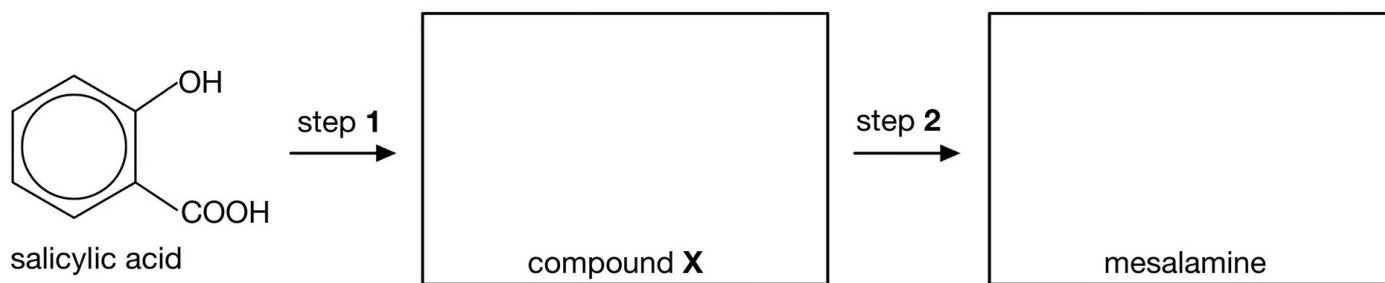


Fig. 3.2

Draw the structures of compound X and mesalamine.

[2 marks]

Question 3c

State the reagents and conditions required for steps 1 and 2 in Fig. 3.2 in part (b).

- step 1

reagent(s)

conditions(s)

- step 2

reagent(s)

conditions(s)

[5 marks]

Model Answers: Hard

1a

a) The difference in bromination of benzene and phenol are:

- Phenol has a lone pair (of electrons) on the oxygen atom

OR

The lone pair of electrons is (partially) delocalised into a benzene ring; [1 mark]

- Benzene electrons / π -bonds are delocalised; [1 mark]
- Benzene has a lower electron density

OR

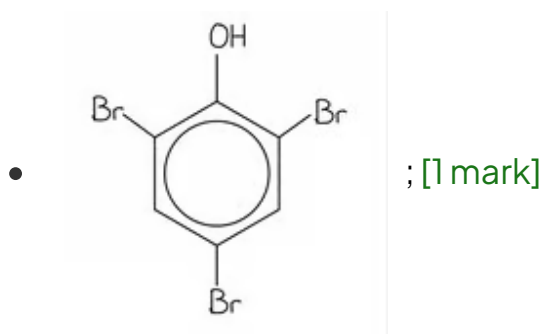
Phenol has a higher electron density; [1 mark]

- Benzene cannot polarise or induce a dipole in the Br_2 / $\text{Br}-\text{Br}$ bond

OR

Phenol can polarise the Br_2 / $\text{Br}-\text{Br}$ bond (allow δ^+ Br or electrophile Br^+ as alternative to polarise); [1 mark]

- Therefore, the reaction of benzene requires a halogen carrier / FeBr_3 / AlBr_3 ; [1 mark]
- Phenol can react with bromine at room temperature to form 2,4,6-tribromophenol; [1 mark]



[Total: 7 marks]

- This question is challenging as there is a lot of information required in one part
- Therefore, break up your answer into smaller parts
 - Bromination of benzene
 - Bromination of phenol

- Difference in reaction
- Structure of phenol when it reacts with bromine
 - Think of the directing effects of side groups / chains when drawing aromatic compounds
- The addition of the oxygen atom in phenol means that phenols react more readily than with electrophiles than compared to benzene
 - In fact, this reaction will happen at room temperature
- One of the electron pairs in the p orbital of oxygen overlaps with the delocalised ring
 - This **increases the electron density of the ring** which means that phenol can polarise the bromine molecule

1b

b)

i) The conditions for reaction 1 are:

- Room temperature
- AND**
- Dilute nitric acid; [1 mark]

ii) To calculate the mass of phenol required:

- M_r 2-nitrophenol = 139.0
- AND**
- M_r of phenol = 94.0; [1 mark]
- Expected mass of 2-nitrophenol = $\frac{30}{35} \times 100 = 85.714$ g
 - Moles of 2-nitrophenol = $\frac{85.714}{139.0} = 0.617$ moles; [1 mark]
 - Therefore, mass of phenol = $0.617 \times 94.0 = 58.0$ g; [1 mark]

[Total: 4 marks]

- The tricky part of the calculation here is determining the mass of 2-nitrophenol assuming a 100% yield
 - It is essentially a backwards % yield calculation

1c

c) Phenol behaves as a weak acid because:

- Hydrogen ion / H^+ can be donated forming phenoxide ion; [1 mark]
- Electronegative O (in phenoxide ion) draws electrons towards it; [1 mark]
- Negative charge on O (in phenoxide ion) will attract hydrogen / H^+ ions back to it; [1 mark]

[Total: 3 marks]

- The pH of a solution of phenol tends to be around 5–6 which means it is a weak acid
- You should be able to:
 - Explain why phenol is a weak acid
 - Give the formula of the phenoxide ion, $C_6H_5O^-$
 - Show how phenol dissociates, $C_6H_5OH \rightleftharpoons C_6H_5O^- + H^+$
- From this, you could also be asked to calculate the pH of a solution of phenol from a given concentration or vice versa!

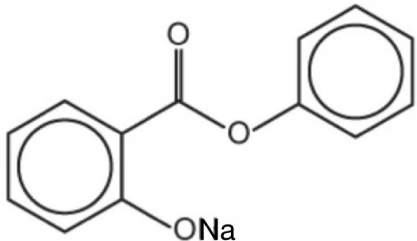
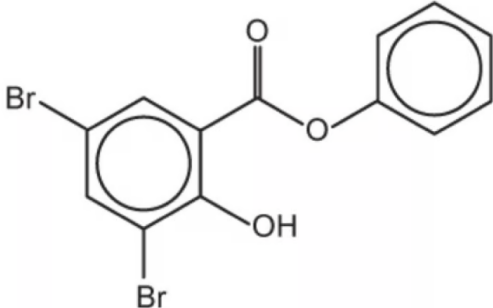
2a

a)

i) The correct number of hybridised carbons are:

- $sp = 0$
AND
 $sp^2 = 13$
AND
 $sp^3 = 0$; [1 mark]

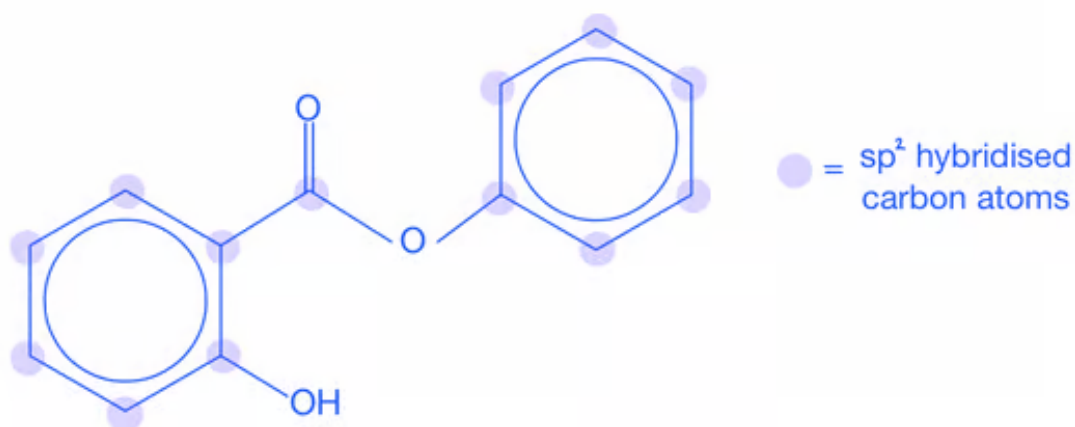
ii) The products and types of reaction are:

Reagent	Structure of product (s)	Type of reaction
Na		Acid-base reaction
Excess Br ₂ (aq)		Bromination / (electrophilic) substitution

- Each correct structure; [1 mark]
- Both correct reaction types; [1 mark]

[Total: 4 marks]

- For part (i)
 - It can be helpful to highlight each carbon and then decide
 - In benzene, the carbon atoms are all sp² hybridised
 - In a carbonyl group C=O, the carbon atoms are sp² hybridised as well



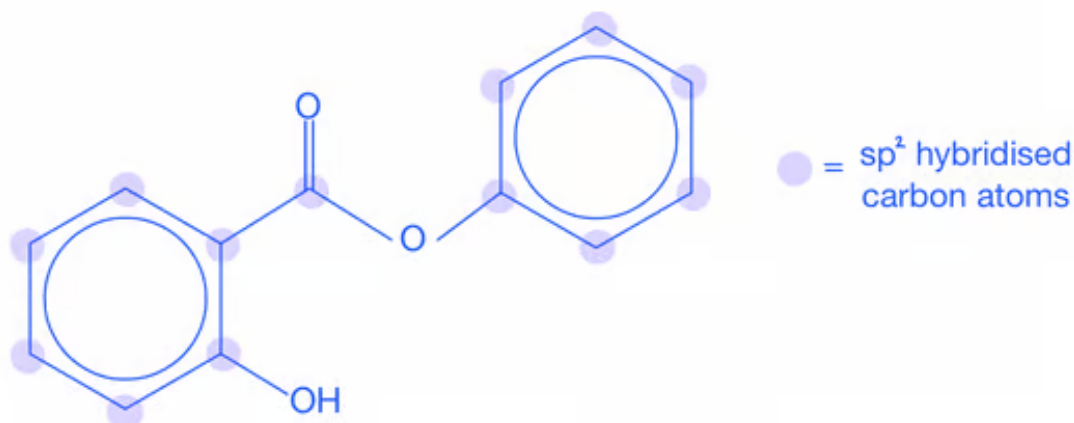
- For part (ii)
 - You need to make sure that you are identifying the correct functional group

that will react with the given reagents

- Both correct reaction types; [1 mark]

[Total: 4 marks]

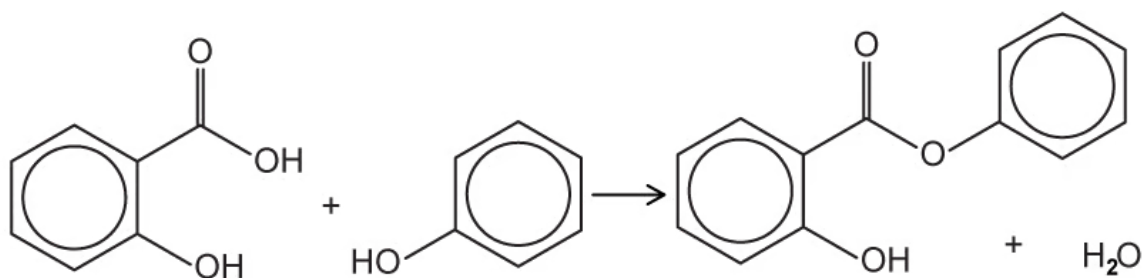
- For part (i)
 - It can be helpful to highlight each carbon and then decide
 - In benzene, the carbon atoms are all sp^2 hybridised
 - In a carbonyl group $C=O$, the carbon atoms are sp^2 hybridised as well



- For part (ii)
 - You need to make sure that you are identifying the correct functional group that will react with the given reagents
 - Excess bromine (aq) will not react with a benzene ring, but it will react with the phenol group
 - The $-OH$ in the phenol group will also react with very reactive metals such as sodium in an acid-base reaction

2b

b) The equation is:



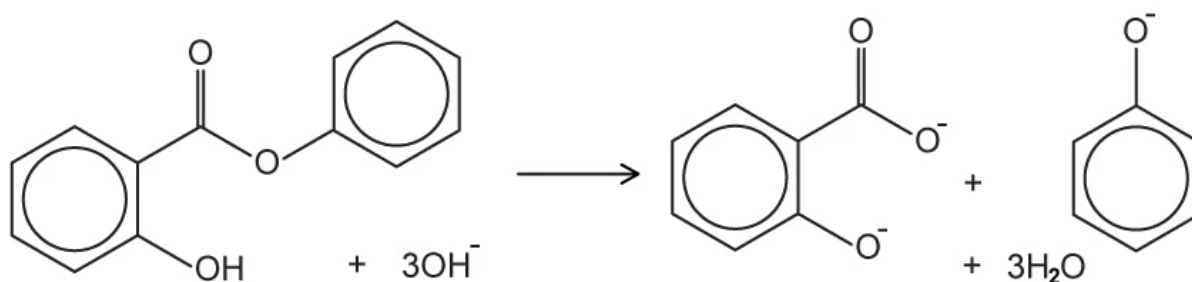
- Correct reactants; [1 mark]
- Correct products; [1 mark]

[Total: 2 marks]

- Notice you are asked to give the displayed formula for this reaction
 - If you are asked for the **fully** displayed formula you must show all the bonds
- A reaction between an alcohol and carboxylic acid is esterification / condensation
 - The small molecule eliminated during the esterification / condensation reaction is water

2c

c) The completed reaction is:



- Hydroxybenzoate **AND** phenol as products; [1 mark]
- Ionisation of at least one of the phenol groups; [1 mark]
- All correct **AND** balanced; [1 mark]

[Total: 3 marks]

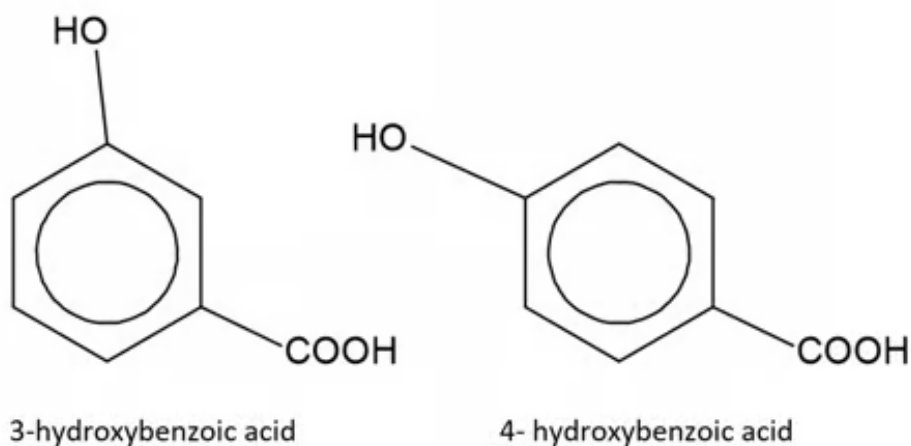
- Phenols will dissolve in alkaline solutions and undergo acid-base reactions with bases to form a soluble salt and water
- The COO^- group will also react with NaOH in alkaline hydrolysis to form a carboxylate salt

3a

a) The systematic IUPAC name for salicylic acid and positional isomers are:

- 2-hydroxybenzoic acid; [1 mark]

There are two other positional isomers:



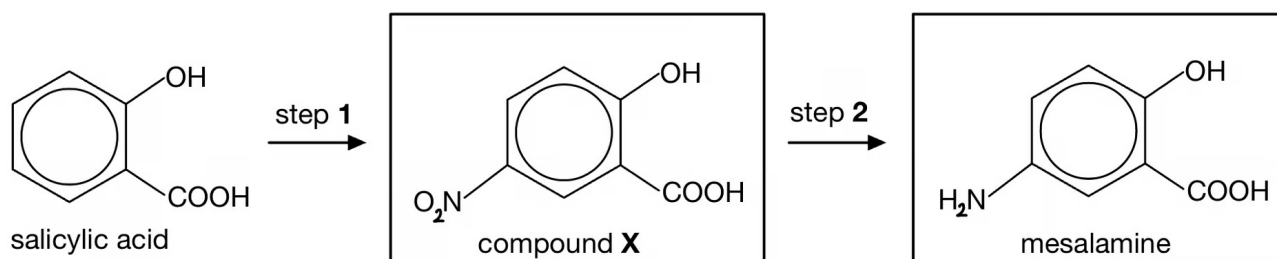
- Correct name & structure of 3-hydroxybenzoic acid; [1 mark]
- Correct name & structure of 4-hydroxybenzoic acid; [1 mark]

[Total: 3 marks]

- The carboxylic acid takes priority over the hydroxyl in naming rules
 - By default it becomes the number 1 carbon on the ring, so the hydroxyl is number 2
- If you move the hydroxyl around the ring it can only be in two more positions
 - **Careful:** 5-hydroxy and 6-hydroxy would be the same as 3-hydroxy and 2-hydroxy, respectively, when counting from the opposite direction

3b

b) The structures of compound **X** and mesalamine are:



- Correct structure of compound **X**; [1 mark]
- Correct structure of mesalamine; [1 mark]

[Total: 2 marks]

- The first part of the question tells you the systematic name for mesalamine, 5-aminosalicylic acid, so you can work out its structure
- The intermediate, compound **X**, links salicylic acid with an aromatic amine
 - So, the addition of a nitro group in the same position as the amine on the mesalamine ring must be the new feature of compound **X**

3c

c)

i) The reagents and conditions for step **1** are:

- Dilute HNO_3 ; [1 mark]
- Room temperature; [1 mark]

ii) The reagents and conditions for step **2** are:

- Sn / HCl ; [1 mark]
- Heat under reflux; [1 mark]
- Excess $\text{NaOH} (\text{aq})$; [1 mark]

[Total: 5 marks]

- The nitration of salicylic acid is not to be confused with the nitration of benzene
 - Salicylic acid is more similar to phenol than benzene
- The presence of an $-\text{OH}$ group is said to be ring activating
 - This increases the electron density in the ring and makes it more attractive to electrophiles
 - This results in the reaction conditions being much milder than the nitration of benzene
- However, the carboxylic acid group in the ring is a deactivating group
 - This reduces the electron density in the ring so will cancel out some of the effect of the $-\text{OH}$ group
- The conversion of compound **X** into mesalamine is a reduction reaction in two steps
 - Without the addition of the NaOH in the second step, the amino group would remain protonated in the acidic conditions as $-\text{NH}_3^+$ instead of NH_2

